SVKM's NMIMS Mukesh Patel School of Technology Management and Engineering, Mumbai

Business Plan for:

Tejas Labs

An educational manufacturing startup focused on designing and producing hands-on STEM

TEJAS LABS

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1. Executive Summary

Tejas Labs is an educational manufacturing startup focused on designing and producing hands-on STEM (Science, Technology, Engineering, and Mathematics) engineering kits for students in India. The company aims to transform traditional science learning into an engaging and practical experience by enabling students to build functional engineering systems through structured DIY kits.

In many education systems, including those prevalent across India, science education often emphasizes theoretical learning and examination performance rather than practical application. This creates a gap between conceptual understanding and real-world problem-solving skills. Tejas Labs seeks to address this gap by providing carefully designed engineering kits that allow students to experiment, construct, and explore scientific principles through hands-on building activities.

Each Tejas Labs kit contains pre-engineered components, assembly tools, and detailed learning guides that enable students to construct working models such as mechanical vehicles, robotic arms, renewable energy systems, and automated devices. By building these systems themselves, students gain a deeper understanding of engineering concepts including mechanics, energy systems, electronics, and automation.

The primary target market includes students aged 10 to 18, parents seeking high-quality educational tools, and schools that are increasingly adopting experiential STEM learning methods. The company will initially focus on the Indian market, where growing awareness of STEM education and innovation-driven learning has created strong demand for practical educational products.

Tejas Labs will generate revenue through direct online sales of engineering kits, subscription-based monthly learning boxes, and bulk partnerships with schools and educational institutions. Distribution will primarily occur through an e-commerce platform supported by digital marketing and institutional collaborations.

The company is currently bootstrapped by its founders, allowing it to maintain operational independence during the early development stage while focusing on product innovation and market validation. Initial funding has been directed toward product design, prototype development, and establishing supplier partnerships.

Over the next five years, Tejas Labs aims to establish itself as a leading provider of affordable and innovative engineering learning kits in India. The company plans to expand its product portfolio, build partnerships with educational institutions, and scale its distribution network across major urban and semi-urban markets.

By combining manufacturing capabilities with educational innovation, Tejas Labs aims to inspire the next generation of engineers, innovators, and problem solvers.

2. Company and Product Description / Business Description

2.1 Business Idea

Tejas Labs is a startup focused on designing and manufacturing **DIY STEM engineering kits** that enable students to learn scientific and engineering concepts through hands-on experimentation and construction.

The core philosophy of the company is **"Learning by Building."**

Our Mission

> *"To democratize high-end engineering education by providing accessible, professional-grade hardware kits and a curriculum that emphasizes creation over consumption."*

Our Vision

> "A world where every student has the tools and confidence to build the next generation of technology, bridging the gap between imagination and reality."

Traditional science education often presents complex engineering and scientific concepts in abstract form. Tejas Labs simplifies these concepts by allowing students to physically assemble working systems, thereby transforming theoretical learning into practical exploration.

Each kit enables students to construct a functional engineering model while simultaneously learning the underlying scientific principles through guided instructions and structured learning materials.

The goal is to cultivate curiosity, creativity, and problem-solving skills among young learners by encouraging them to experiment with real engineering systems.

2.2 Founders and Leadership

Tejas Labs was founded by **Dev Vora** and **Siddharth Vaddepalli**, both technology enthusiasts with a strong interest in innovation, engineering, and education.

Dev Vora – Chief Executive Officer (CEO):

Dev Vora is responsible for the strategic vision and overall direction of the company. As CEO, he oversees product innovation, business strategy, partnerships with educational institutions, and long-term growth planning. His background in technology and analytics enables the company to design learning solutions that integrate engineering, creativity, and modern educational methods.

LinkedIn: https://www.linkedin.com/in/vora-dev

Siddharth Vaddepalli – Chief Financial Officer (CFO):

Siddharth Vaddepalli leads the financial and operational management of Tejas Labs. As CFO, he is responsible for financial planning, budgeting, cost optimization, and operational scalability. His role ensures that the company maintains financial sustainability while expanding its manufacturing and distribution capabilities.

The founders combine technical understanding with strategic and financial management skills, enabling Tejas Labs to develop innovative products while maintaining strong operational discipline.

LinkedIn: https://www.linkedin.com/in/siddharth-vaddepalli

2.3 Current Funding Status

Tejas Labs is currently a **33% bootstrapped startup with a Loan for the rest 66%**, meaning the company is funded entirely through the founders' personal investment and internal resources during its early development stage.

Bootstrapping provides several advantages during the initial phase:

- Full ownership and strategic control by the founders
- Flexibility to focus on product development and market validation
- Reduced dependency on external investors during the early stages

Initial funding has been directed toward:

- Product design and prototype development
- Component sourcing and supplier identification
- Initial kit assembly and testing
- Brand identity development and website creation

As the company gains market traction and expands its product portfolio, Tejas Labs may explore external funding options such as angel investors, startup incubators, or educational innovation grants.

2.4 Product Description

Tejas Labs manufactures a range of engineering kits designed to introduce students to core STEM concepts through hands-on building experiences. Each kit focuses on a specific engineering principle and enables students to construct a functioning mechanical or electronic system.

Each kit contains:

- Pre-designed mechanical components
- Electronic components such as motors, sensors, and wiring
- Assembly tools when required
- Illustrated step-by-step instruction manuals
- Access to digital tutorials and guided learning resources

Example product offerings include:

- **Mechanical Motion Kit**
  Students build a rubber-band powered vehicle to understand energy storage, motion, and mechanical efficiency.
- **Hydraulic Robotic Arm Kit**
  Students construct a robotic arm that demonstrates fluid mechanics and force transmission through hydraulic systems.
- **Renewable Energy Exploration Kit**
  Students assemble a working wind turbine or solar-powered system to understand sustainable energy generation.
- **Automation and Electronics Kit**
  Students build a basic automated device that introduces electronic circuits and control mechanisms.

The product portfolio is designed with progressive levels of complexity so that students can advance from beginner kits to more sophisticated engineering builds.

2.5 Logo and Brand Identity

The Tejas Labs logo reflects the company's mission of making science and engineering learning engaging and accessible for students.

The logo features the **company name "Tejas Labs" written in a playful and approachable font**, symbolizing creativity, curiosity, and the excitement of discovery. The informal and energetic typography helps make the brand appealing to younger audiences while reinforcing the idea that learning engineering can be fun and interactive.

Positioned above the company name is a **laboratory beaker**, which represents experimentation, scientific exploration, and innovation. The beaker symbolizes the spirit of experimentation that lies at the core of STEM education and reflects the company's focus on hands-on learning and discovery.

Together, the playful typography and the laboratory beaker communicate the brand's identity as a creative, student-friendly STEM innovation company that encourages experimentation, building, and curiosity.

2.6 Innovation and Use of Design Thinking

Tejas Labs adopts the **design thinking framework** to ensure that its products effectively address real educational challenges faced by students and educators.

- **Empathize**
  Understand how students struggle to visualize and apply scientific concepts through traditional theoretical learning.
- **Define**
  Identify the gap between textbook-based education and practical engineering understanding.
- **Ideate**
  Develop interactive engineering kits that simplify complex concepts through hands-on experimentation.
- **Prototype**
  Design and test early versions of kits with students to evaluate usability and educational effectiveness.
- **Test and Refine**
  Continuously improve product design based on feedback from users and educators.

This iterative process ensures that each kit remains engaging, educational, and user-friendly.

2.7 Points of Differentiation

Tejas Labs differentiates itself from conventional educational toys and science kits through several key advantages.

- **Hands-on Engineering Learning**
  Students construct working mechanical and electronic systems rather than simple demonstration toys.
- **Structured Educational Design**
  Each kit integrates learning objectives that explain the scientific principles behind the build process.
- **Progressive Learning Ecosystem**
  Students can progress from beginner kits to advanced engineering projects.
- **Integration with Digital Learning**
  Online tutorials and instructional videos complement the physical building experience.

- **Affordability for the Indian Market**
  Products are designed to remain accessible for middle-income households and educational institutions.

2.8 Business Model Canvas

Component	Description
Customer Segments	Students aged 10–18, parents, schools, STEM education centers
Value Proposition	Hands-on engineering learning through interactive DIY STEM kits
Channels	Company website, educational partnerships, online marketplaces
Customer Relationships	Online support, tutorials, community engagement
Revenue Streams	Individual kit sales, subscription STEM boxes, institutional bulk sales
Key Resources	Product designs, engineering expertise, supplier network, assembly facilities
Key Activities	Product development, component sourcing, kit assembly, quality control, marketing
Key Partners	Component suppliers, packaging manufacturers, logistics providers, educational institutions
Cost Structure	Raw materials, manufacturing labor, packaging, logistics, marketing, research and development

3. Industry Analysis

3.1 Industry Overview

Tejas Labs operates within the intersection of three major and rapidly evolving sectors: the educational toys industry, the STEM education market, and experiential learning products. Over the past decade, these sectors have experienced strong growth as educational institutions and parents increasingly prioritize practical learning, creativity, and problem-solving skills in children.

Traditional classroom education in many countries, including India, has historically focused heavily on theoretical instruction and examination performance. However, the rapid advancement of technology-driven industries has created growing demand for skills such as analytical thinking, engineering literacy, and innovation. As a result, there has been a shift toward experiential learning models, where students learn scientific and engineering concepts through experimentation and project-based activities.

Hands-on STEM learning tools such as robotics kits, engineering construction sets, and science experiment kits have become increasingly popular as they help bridge the gap between theoretical learning and real-world application. Companies operating in this space design products that encourage students to explore scientific principles through building, experimentation, and creative problem solving.

Tejas Labs aims to position itself within this ecosystem as a manufacturer of affordable and structured engineering learning kits designed specifically for the Indian education market.

3.2 Industry Size

The educational toys and STEM learning market has grown rapidly in recent years due to increasing awareness of skill-based education.

According to market research data frequently cited by financial intelligence platforms such as Bloomberg and industry research firms, the global STEM toys market is projected to reach approximately USD 1.83 billion by 2030, driven by growing demand for products that promote science and engineering skills among children.

In India, the educational toy sector has also expanded significantly. Market research from Grand View Research estimates that the Indian educational toys market was valued at approximately USD 4.41 billion in 2023 and is projected to reach around USD 8.86 billion by 2030, reflecting increasing consumer demand for learning-based toys and educational products.

Within this industry, the technology-enabled toy segment in India is estimated to be worth approximately USD 1.6 billion in 2024, with increasing adoption driven by the rise of STEM education programs and growing digital awareness among parents.

These figures indicate that the Indian market presents significant opportunities for startups that develop innovative and educational STEM products.

3.3 Industry Growth Rate

The STEM education sector continues to grow rapidly worldwide due to increasing emphasis on innovation-driven economies.

The Indian educational toy market is projected to grow at a compound annual growth rate (CAGR) of approximately 10.5 percent between 2024 and 2030.

The technology-enabled toy segment in India is expected to grow even faster at approximately 14.7 percent CAGR, driven by rising interest in robotics, coding, and engineering learning tools.

Additionally, the K–12 STEM education market in India generated approximately USD 2.76 billion in 2024 and is expected to reach over USD 8.27 billion by 2030, reflecting an annual growth rate of nearly 19.7 percent.

These growth projections demonstrate strong demand for educational products that combine learning with interactive experimentation.

3.4 Industry Structure

The STEM learning industry includes several categories of participants:

- **Educational toy manufacturers**
  Companies producing science kits, building toys, and educational games.
- **STEM learning startups**
  Companies focused on robotics kits, coding tools, and engineering learning systems.
- **EdTech companies**
  Platforms offering digital learning solutions, coding courses, and online STEM education.
- **Educational institutions and curriculum developers**
  Organizations integrating STEM learning tools into school programs.

Many companies in the sector are increasingly combining physical learning kits with digital learning platforms to provide a more comprehensive educational experience.

While large global companies dominate certain segments, the Indian market continues to offer opportunities for locally designed products that are affordable and aligned with domestic education systems.

3.5 Nature of Industry Participants

Participants in this industry range from large multinational toy manufacturers to small innovation-driven startups.

Large toy companies operate through large-scale manufacturing and retail distribution networks. These firms typically focus on mass-market educational toys.

EdTech companies focus on digital learning platforms that teach coding, robotics, and technology skills.

Startup companies often specialize in niche areas such as robotics kits, maker education tools, or hands-on engineering kits designed to provide deeper learning experiences.

Tejas Labs will position itself as a specialized manufacturer of engineering learning kits that combine physical experimentation with structured educational guidance.

3.6 PESTLE Analysis of the STEM Education Industry in India

A PESTLE analysis helps evaluate the macro-environmental factors that influence the industry.

3.6.1 Political Factors

The Indian government has increasingly promoted STEM education and innovation through national initiatives such as:

- Make in India
- Atal Innovation Mission
- Startup India
- National Education Policy 2020

These initiatives encourage innovation, technology education, and domestic manufacturing, which supports the growth of companies like Tejas Labs.

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3.6.2 Economic Factors

India has a rapidly growing middle class with increasing disposable income and a strong focus on education. Parents are willing to invest in educational products that improve their children's academic and technical skills.

Additionally, the expansion of the Indian toy manufacturing sector creates opportunities for domestic startups to compete with imported educational toys.

3.6.3 Social Factors

Parents and educators are increasingly emphasizing skill-based learning rather than rote memorization. There is growing awareness of the importance of STEM education in preparing students for future careers in technology and engineering.

Hands-on learning tools are becoming popular because they make education more engaging and interactive.

3.6.4 Technological Factors

Advances in electronics, sensors, microcontrollers, and digital learning platforms have made it easier for companies to design innovative educational kits.

Integration of digital tutorials, online communities, and video-based learning has further enhanced the effectiveness of STEM learning tools.

3.6.5 Legal Factors

Companies operating in the educational toy industry must comply with product safety standards, manufacturing regulations, and intellectual property laws.

Ensuring product safety, especially for children's products, is critical to maintaining trust and credibility.

3.6.6 Environmental Factors

There is increasing emphasis on sustainable manufacturing practices and environmentally friendly materials.

Companies that adopt eco-friendly packaging and responsible sourcing can strengthen their brand reputation among environmentally conscious consumers.

3.7 Porter's Five Forces Analysis

3.7.1 Threat of New Entrants – Moderate

While the startup ecosystem encourages innovation, developing high-quality educational products requires strong design capabilities, manufacturing partnerships, and brand credibility. These factors create moderate barriers to entry.

3.7.2 Bargaining Power of Suppliers – Moderate

Suppliers of electronic components, motors, and mechanical parts may influence production costs. However, the availability of multiple suppliers reduces dependency on any single vendor.

3.7.3 Bargaining Power of Buyers – High

Customers have access to many educational products and alternatives. Parents and schools often compare price, educational value, and brand credibility before making purchasing decisions.

3.7.4 Threat of Substitutes – Moderate

Alternative learning solutions such as online STEM courses, educational apps, and robotics workshops may substitute for physical learning kits. However, hands-on building experiences offer unique educational benefits that digital platforms alone cannot replicate.

3.7.5 Industry Rivalry – Increasing

The STEM learning market is becoming more competitive as startups, toy manufacturers, and edtech companies expand their offerings. Innovation, product quality, and strong branding will be critical for differentiation.

3.8 Key Success Factors

Companies operating in this industry must focus on several critical success factors.

- Product innovation and engaging design
- Strong educational value and curriculum relevance
- Affordable pricing for wider market access
- High product quality and safety standards
- Effective digital marketing and distribution channels
- Strong brand credibility among parents and schools

3.9 Long-Term Industry Prospects

The long-term outlook for the STEM education industry is highly positive.

As economies become increasingly driven by technology and innovation, demand for engineers, scientists, and technical professionals will continue to grow. Governments and educational institutions are therefore placing greater emphasis on developing STEM capabilities among students at an early age.

India, with its large student population and strong focus on engineering education, represents one of the largest potential markets for STEM learning tools.

Companies that successfully combine innovation, affordability, and educational value will be well positioned to scale within this rapidly expanding market.

4. Market Analysis

4.1 Market Overview

The market for STEM-based educational products in India has grown rapidly in response to increasing awareness of the importance of experiential learning and early exposure to science and engineering concepts. As industries worldwide transition toward technology-driven economies, the demand for problem-solving, analytical thinking, and innovation skills has significantly increased. This shift has encouraged parents, educators, and policymakers to prioritize STEM education as a foundational element of modern learning.

India represents one of the largest potential markets for STEM learning tools due to its large school-age population and strong emphasis on technical education. With over **250 million school students**, India has one of the largest K–12 education systems globally. Within this ecosystem, parents and schools are increasingly investing in supplementary learning tools that provide practical exposure beyond textbook-based education.

Hands-on educational products such as engineering kits, robotics tools, and science experiment sets are gaining popularity as they enable students to explore complex concepts through interactive learning. These products support the development of curiosity, creativity, and critical thinking while making scientific learning more engaging.

Tejas Labs aims to capitalize on this growing demand by providing structured engineering learning kits that combine **hands-on experimentation with guided educational content**, specifically tailored for the Indian education ecosystem.

4.2 Market Segmentation

A structured segmentation strategy allows Tejas Labs to identify and prioritize the most promising customer groups within the broader STEM education market.

4.2.1 Demographic Segmentation

The primary demographic segment consists of **school students between the ages of 10 and 16**, a stage where students begin to develop interest in science, engineering, and technology-related subjects.

Students aged **10–12 years** represent the introductory segment, where basic engineering concepts such as mechanical motion, energy systems, and simple machines can be introduced through guided experimentation.

Students aged **13–16 years** represent a more advanced segment where kits can incorporate deeper engineering principles such as electronics, structural design, automation, and problem-solving challenges.

Parents within the **middle-income and upper-middle-income households** form the key purchasing decision-makers within this segment, as they typically invest in supplementary educational resources for their children.

4.2.2 Institutional Segmentation

Educational institutions represent an important secondary market segment.

This segment includes:

- Private schools implementing STEM education programs
- Innovation labs and maker spaces within schools
- After-school learning centers offering robotics or coding classes
- STEM-focused educational workshops and summer programs

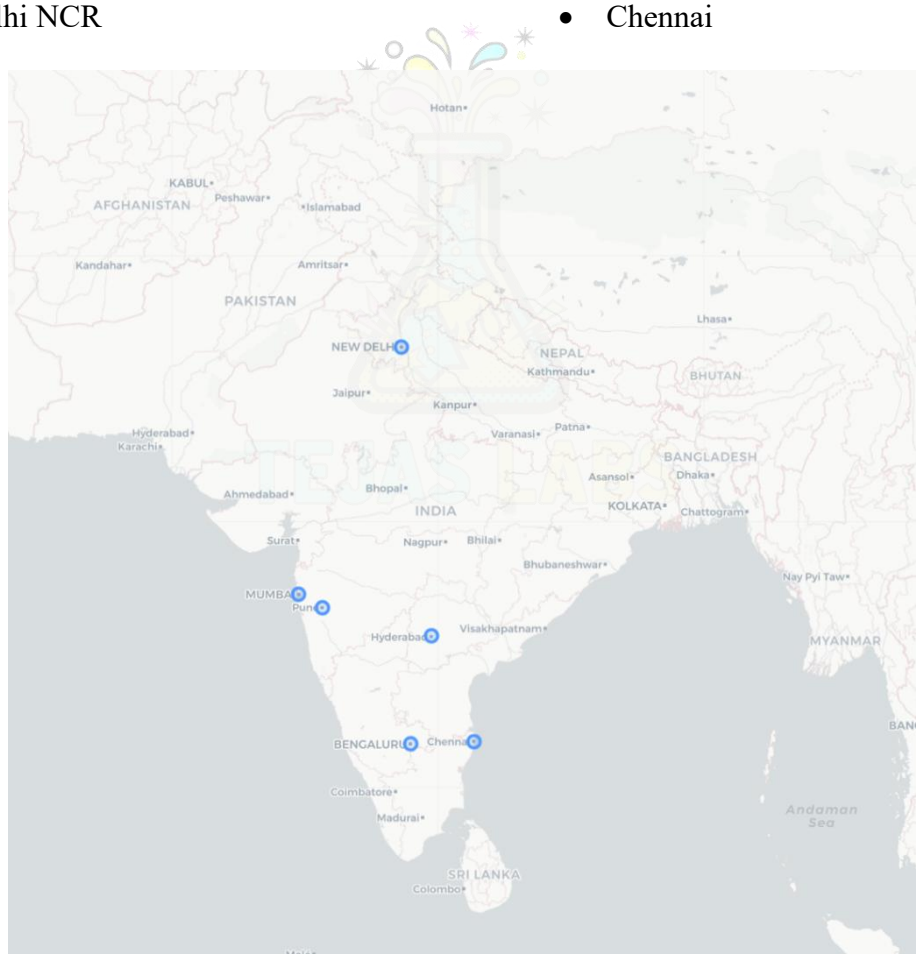
These institutions frequently purchase educational kits in bulk and use them as part of structured learning programs.

4.2.3 Geographic Segmentation

The initial focus will be on **Tier 1 and Tier 2 urban markets**, where awareness of STEM education and purchasing power are relatively high.

Key geographic markets include major metropolitan regions such as:

- Mumbai
- Bangalore
- Delhi NCR
- Pune
- Hyderabad
- Chennai



These cities have a high concentration of private schools, technology-driven families, and innovation-focused educational programs, making them ideal early adoption markets.

4.3 Target Market Selection

Based on the segmentation analysis, Tejas Labs will adopt a **dual-target market strategy** focusing on both direct consumer sales and institutional partnerships.

4.3.1 Primary Target Market

The primary target market consists of **parents of students aged 10–16 years** who actively seek educational tools that enhance learning outcomes and stimulate intellectual curiosity.

This segment typically exhibits the following characteristics:

- High emphasis on academic performance and skill development
- Strong interest in STEM education and future career opportunities
- Willingness to invest in educational resources that provide long-term learning benefits

Parents in metropolitan and technology-oriented cities are particularly receptive to innovative learning tools.

4.3.2 Secondary Target Market

The secondary target market includes **educational institutions and STEM learning centers**.

Schools increasingly integrate experiential learning programs into their curricula through:

- Science fairs
- Innovation labs
- Project-based learning initiatives
- Robotics and maker clubs

Tejas Labs will offer specialized educational kits and structured learning modules designed for classroom use, allowing schools to integrate engineering learning into existing science programs.

4.4 Buyer Behavior Analysis

Understanding purchasing motivations and decision-making patterns is critical for developing an effective marketing strategy.

4.4.1 Parent Purchasing Behavior

Parents typically evaluate educational products using several key criteria:

Educational value and skill development potential

- Product safety and durability
- Age-appropriate design and instructions
- Brand reputation and peer recommendations
- Price affordability relative to perceived learning benefits

Parents increasingly rely on online reviews, educational influencers, and digital content when evaluating educational products.

4.4.2 Institutional Purchasing Behavior

Educational institutions follow a more structured procurement process.

Schools typically evaluate STEM learning tools based on:

- Alignment with curriculum and learning objectives
- Ease of implementation in classroom settings
- Durability and product safety standards
- Availability of teaching guides and structured activity modules
- Cost efficiency for bulk purchases

Establishing credibility with educators and administrators is therefore essential for institutional sales.

4.5 Competitive Landscape

The competitive environment within the STEM educational products industry includes several categories of competitors.

4.5.1 Educational Toy Manufacturers

Traditional educational toy companies produce science kits, building toys, and educational games designed primarily for entertainment-based learning.

While these products often introduce scientific concepts, they may lack structured educational frameworks or deeper engineering challenges.

4.5.2 EdTech Platforms

Digital learning platforms offer online coding courses, virtual STEM learning programs, and interactive educational content.

Although these platforms provide valuable theoretical instruction, they do not fully replace the learning experience derived from **physical experimentation and hands-on construction**.

4.5.3 STEM Learning Startups

A growing number of startups focus specifically on robotics kits, coding devices, and maker education tools.

These companies often target premium market segments and emphasize advanced technological features.

However, their pricing structures may limit accessibility for broader segments of the Indian market.

4.5.4 Competitive Matrix

To clearly illustrate Tejas Labs' positioning in the Indian STEM educational kit market, the following competitive matrix compares key players across price, strengths, target segments, and Tejas Labs' differentiation.

Competitive Matrix

Competitor	Price Range (₹)	Core Strength	Primary Target	Tejas Labs Differentiation
Smartivity	500 – 2,000	Subscription-based mechanical & STEAM kits	Ages 6–12, parents seeking fun DIY	Deeper engineering focus (hydraulics, automation) with curriculum-aligned guides
Butterfly EduFields	800 – 3,599	Robotics & electronics kits with school projects	Ages 8–14, schools & after-school	Progressive tiered kits at 20–40% lower price + "Learning by Building" philosophy
Robocraze	1,000 – 5,000	Affordable robotics & coding kits	Ages 10–16, coding enthusiasts	Broader mechanical + hydraulic systems with India-specific curriculum alignment

Competitor	Price Range (₹)	Core Strength	Primary Target	Tejas Labs Differentiation
STEMpedia	1,500 – 6,000	AI, IoT & advanced electronics kits	Schools, innovation labs, ages 12+	More accessible pricing for parents + hands-on physical building
LEGO Education	2,000 – 8,000+	Global brand with excellent app integration	Premium segment, affluent parents	60–70% lower price + full focus on NEP 2020 and Make-in-India components

This matrix highlights Tejas Labs' unique positioning in the **affordable, hands-on engineering niche** that combines mechanical depth, structured educational content, and India-specific accessibility.

4.6 Competitive Positioning of Tejas Labs

Tejas Labs will differentiate itself through a combination of affordability, structured learning design, and engaging engineering challenges.

The company's core value proposition will focus on three pillars:

- Accessible pricing suitable for the Indian consumer market
- Structured engineering projects that teach real scientific principles
- Engaging product design that encourages curiosity and experimentation

This positioning enables Tejas Labs to occupy a niche between entertainment-focused educational toys and high-cost robotics platforms.

4.7 Go-To-Market Strategy

To successfully establish market presence, Tejas Labs will adopt a phased market entry strategy.

4.7.1 Phase 1: Direct-to-Consumer Digital Launch

The initial launch will occur through the company's website and online marketing channels.

Digital marketing efforts will include:

Social media campaigns targeting parents interested in STEM education
Video demonstrations showcasing engineering projects
Collaborations with educational content creators and influencers

4.7.2 Phase 2: Institutional Partnerships

Following initial brand development, Tejas Labs will expand into the institutional segment through partnerships with schools and learning centers.

The company will offer:

- Bulk purchasing packages
- Teacher guides and classroom activity modules
- STEM workshop kits for school innovation programs

4.7.3 Phase 3: Community Engagement and Workshops

Hands-on workshops and science demonstration events will play a critical role in building brand awareness and product credibility.

These events will allow students to experience engineering kits directly and encourage adoption among both parents and educators.

4.8 Sales Forecast and Market Share

During the early growth stages, Tejas Labs will prioritize brand awareness and product refinement while gradually scaling production capacity.

4.8.1 2026

Focus on product launch and brand establishment through online channels.

Estimated units sold: **3,000–5,000 kits**

Estimated revenue: **₹30–40 lakhs**

4.8.2 2028

Expansion into school partnerships and wider geographic distribution.

Estimated units sold: **15,000–20,000 kits annually**

Estimated revenue: **₹2–3 crores**

4.8.3 2030

Establishment of Tejas Labs as a recognized STEM learning brand within the Indian education ecosystem.

Estimated units sold: **40,000–50,000 kits annually**

Estimated revenue: **₹7–10 crores**

4.9 Market Opportunity

Given the size of the Indian student population and the increasing demand for experiential learning tools, the long-term growth potential for STEM education products remains significant.

Companies that successfully combine innovative product design, educational credibility, and accessible pricing will be well positioned to capture substantial market share within this rapidly expanding sector.

Tejas Labs aims to leverage these opportunities by establishing itself as a trusted provider of engaging engineering learning experiences for students across India.

5. Marketing Plan

5.1 Strategic Marketing Objective

The primary marketing objective of Tejas Labs is to establish the brand as a credible and engaging STEM learning platform that enables students to experience engineering concepts through hands-on experimentation. The marketing strategy is designed to support three key growth goals during the initial years of operation:

1. Build strong brand awareness within the STEM education ecosystem in India
2. Establish educational credibility among parents, educators, and schools
3. Drive scalable product adoption through both direct consumer channels and institutional partnerships

Given the company's initial bootstrapped funding structure, marketing investments will focus on high-impact, cost-efficient digital and experiential channels rather than traditional mass

advertising. The strategy emphasizes content-driven engagement, community building, and educational partnerships to build trust and long-term brand equity.

5.2 Brand Strategy

5.2.1 Brand Identity and Positioning

Tejas Labs will position itself as a hands-on engineering learning brand designed for the next generation of innovators. Unlike traditional educational toys that primarily focus on entertainment or passive learning, Tejas Labs aims to provide structured engineering experiences that allow students to actively build, test, and understand real-world scientific principles.

The brand identity combines three core elements:

- Curiosity
- Experimentation
- Engineering creativity

The visual identity of the company reinforces this positioning. The logo, consisting of the company name rendered in a playful font with a laboratory beaker placed above it, communicates both the scientific foundation and the creative exploration aspect of the brand. The design is intended to resonate with younger audiences while maintaining credibility within the educational ecosystem.

5.2.2 Brand Value Proposition

The value proposition of Tejas Labs is centered around enabling students to learn engineering by building real working models rather than simply reading about scientific concepts.

The brand delivers value to customers through three primary dimensions.

- **Experiential Learning**
  Students gain practical exposure to engineering principles through hands-on experimentation and construction activities.
- **Structured Educational Design**
  Each kit includes detailed instructions, explanatory guides, and engineering challenges that transform experimentation into meaningful learning experiences.
- **Accessibility**
  Products are designed and priced to be accessible to Indian households while maintaining high educational value.

5.3 Target Customer Strategy

The marketing strategy for Tejas Labs is built around two primary customer segments.

5.3.1 Direct Consumer Segment

The primary customer segment consists of parents of students aged 10–16 years who actively seek educational resources that support skill development beyond traditional classroom learning.

These parents typically belong to middle-income and upper-middle-income urban households and demonstrate strong interest in STEM education, technology exposure, and academic enrichment activities.

For this segment, marketing communication will emphasize:

Development of analytical and problem-solving abilities
Hands-on learning experiences that complement school education
Fun, engaging experimentation that stimulates curiosity

The messaging will focus on positioning Tejas Labs as a product that enables children to explore engineering concepts independently while having an enjoyable learning experience.

5.3.2 Institutional Segment

The secondary customer segment includes schools, innovation labs, and STEM learning centers that integrate project-based learning and experimentation into their teaching methodologies.

These institutions seek tools that can enhance science education through practical demonstrations and structured student projects.

For this segment, the value proposition will emphasize:

Alignment with STEM and innovation-based education programs
Ease of integration into classroom and laboratory activities
Availability of structured teaching guides and learning modules

Institutional adoption will also help build credibility for the brand within the broader educational ecosystem.

5.4 Marketing Channel Strategy

The marketing channel strategy for Tejas Labs is designed to maximize reach while maintaining cost efficiency during the early stages of business growth.

5.4.1 Digital Marketing

Digital marketing will serve as the primary customer acquisition channel due to its scalability and cost efficiency.

Key digital strategies include:

Search engine optimization targeting queries related to STEM learning, science kits, and engineering toys
Targeted social media campaigns focusing on parents interested in educational products
Educational video content demonstrating engineering experiments and kit assembly

Video-based content will play a particularly important role because it allows potential customers to visualize the learning experience and product functionality.

Platforms such as YouTube and Instagram will be leveraged to showcase engaging demonstrations of engineering kits being built and tested.

5.4.2 Content Marketing

Content marketing will form a central pillar of the brand development strategy.

Rather than focusing solely on product promotion, Tejas Labs will position itself as a thought leader in hands-on STEM education.

Content initiatives will include:

- Educational articles explaining engineering concepts
- Video tutorials demonstrating science experiments
- Project challenges encouraging students to innovate and modify existing kits

This strategy not only attracts potential customers but also strengthens brand credibility within the education community.

5.4.3 Community Engagement

A strong community-driven learning environment can significantly enhance customer engagement and brand loyalty.

Tejas Labs will encourage users to share their projects, modifications, and creative experiments through online communities and social media platforms.

Community engagement initiatives may include:

- Student innovation challenges
- Monthly engineering build competitions
- Online showcases of student projects

This participatory approach transforms customers into active members of the Tejas Labs learning ecosystem.

5.4.4 Educational Partnerships

Partnerships with educational institutions will serve as a critical channel for brand expansion.

Tejas Labs will collaborate with schools, STEM clubs, and innovation labs to introduce its kits into structured learning programs.

These partnerships may involve:

- Classroom demonstrations
- STEM workshops for students
- Teacher training sessions on using engineering kits as learning tools

Institutional partnerships help establish the brand as an educational partner rather than simply a product manufacturer.

5.5 Pricing Strategy

Tejas Labs will adopt a **value-based pricing strategy** that reflects the educational value of its products while maintaining accessibility for the Indian consumer market.

The pricing strategy will be structured across multiple product tiers to accommodate different levels of complexity and learning depth.

Entry-level engineering kits will be priced at **₹1,999**, targeting younger students and introductory engineering concepts.

Intermediate kits designed for more complex engineering projects will be priced at **₹3,499**.

Advanced kits involving electronics or multi-stage engineering builds may be priced at **₹6,999**.

This tiered pricing structure allows customers to gradually progress through increasingly sophisticated learning experiences.

For institutional customers, **volume-based pricing discounts** will be offered to encourage bulk purchases and long-term partnerships.

5.6 Distribution Strategy

The distribution strategy will follow a **direct-to-consumer first approach**, supported by digital sales channels and institutional partnerships.

5.6.1 Direct Online Sales

The primary distribution channel will be the official Tejas Labs website, which will function as both an e-commerce platform and an educational resource hub.

The website will feature:

- Product pages with demonstration videos
- Educational blog content
- Step-by-step build guides and experiment tutorials

This integrated approach allows the company to combine product sales with educational engagement.

5.6.2 Marketplace Expansion

After initial traction is established, Tejas Labs may expand its distribution through established e-commerce marketplaces such as Amazon and Flipkart.

These platforms provide access to large customer bases and established logistics networks, which can accelerate product adoption.

5.6.3 Institutional Sales

Institutional sales will follow a **direct engagement model**.

This will involve:

- Approaching schools and innovation labs
- Conducting product demonstrations
- Offering bulk purchasing packages for STEM programs

Although institutional sales cycles are typically longer than consumer sales cycles, they often generate higher order volumes and long-term partnerships.

5.7 Sales Process

The consumer sales journey typically follows a multi-stage process.

Awareness is created through digital marketing content and educational videos. Interested customers visit the Tejas Labs website to explore product offerings. Customers evaluate product benefits through demonstrations, tutorials, and reviews. Purchase decisions are made through the online store.

Post-purchase engagement through tutorials and engineering challenges helps maintain customer interest and encourages repeat purchases.

5.8 Customer Retention and Lifetime Value

Customer retention is particularly important in the STEM learning market because students can progress through increasingly advanced kits over time.

Tejas Labs will encourage repeat purchases through:

- Sequential product series that build upon previous kits
- New engineering challenges and add-on modules
- Ongoing educational content and project ideas

This approach transforms the product line into a learning progression pathway, increasing the lifetime value of each customer.

5.9 Long-Term Marketing Vision

Over time, Tejas Labs aims to evolve from a product manufacturer into a **recognized STEM learning ecosystem**.

The long-term marketing vision includes:

- Developing a large community of young innovators and makers
- Hosting national engineering challenges for students
- Partnering with educational organizations and innovation programs

By combining engaging products with a strong learning community, Tejas Labs can build a brand that extends beyond individual kits and becomes a platform for student innovation.

6. Operations Plan

6.1 Operational Strategy

The operations strategy of Tejas Labs is designed to support the efficient design, manufacturing, and distribution of hands-on STEM engineering kits while maintaining high standards of product quality, educational effectiveness, and cost efficiency. As a manufacturing-oriented startup operating within the educational products industry, operational excellence is essential to ensure that the products remain accessible to the Indian market while maintaining durability and safety standards.

In its early stage, Tejas Labs will adopt a **lean and asset-light operating model** that combines in-house product design and educational development with outsourced manufacturing and assembly partnerships. This approach allows the company to minimize capital expenditure during the early stages while maintaining flexibility in scaling production volumes based on market demand.

The operational structure is designed around three strategic priorities:

- Efficient product development and rapid prototyping
- Scalable manufacturing through contract production partners
- Reliable supply chain and fulfillment infrastructure

This hybrid operating model enables Tejas Labs to focus its internal resources on product innovation, educational design, and brand development, while leveraging the expertise of established manufacturing partners for production and assembly.

6.2 Operational Workflow

The operational workflow of Tejas Labs is organized into five core stages that collectively transform product concepts into finished engineering kits delivered to customers.

1. Product Design and Engineering Development
2. Supplier Procurement and Component Sourcing
3. Manufacturing and Assembly
4. Quality Assurance and Product Testing
5. Distribution and Order Fulfillment

Each stage is structured to ensure operational efficiency, product reliability, and the ability to scale production as the business grows.

6.3 Product Development and Engineering Design

Product innovation is the foundation of the Tejas Labs business model. All engineering kits will be conceptualized and designed internally by the founding team to ensure strong alignment between educational objectives and product functionality.

The product development process will follow a structured **design thinking framework**, allowing the company to design products that are both educationally effective and engaging for students.

The development cycle includes the following stages:

- Identification of engineering concepts to be taught
- Ideation and product concept development
- Engineering design and digital prototyping
- Physical prototype development and testing
- Educational content and instruction manual development
- Final product validation and readiness for production

During the prototyping stage, rapid iteration will be enabled through the use of **3D modeling tools and rapid prototyping technologies**. This approach allows the team to refine product functionality and usability before transitioning to full-scale manufacturing.

Each engineering kit will be designed to achieve three educational objectives:

- Demonstrate real scientific or engineering principles
- Encourage hands-on experimentation and problem-solving
- Provide structured learning through guided instructions and challenges

6.4 Procurement and Supplier Management

The production of engineering kits requires the integration of multiple components including mechanical parts, electronic modules, structural elements, and packaging materials.

Tejas Labs will establish supplier partnerships with manufacturers across key industrial clusters within India. Procuring components domestically provides several operational advantages including reduced logistics costs, faster procurement cycles, and alignment with the government's **Make in India initiative**.

Primary supplier categories will include:

- Mechanical component suppliers for plastic and metal parts
- Electronic component suppliers for motors, wires, sensors, and switches
- Packaging suppliers for product boxes, printed manuals, and inserts
- Printing partners for instructional guides and educational materials

To reduce supply chain risk, the company will maintain multiple approved vendors for critical components, ensuring continuity of production in case of supplier disruptions.

Supplier relationships will be managed through long-term procurement agreements that ensure consistent component quality, competitive pricing, and predictable delivery timelines.

6.5 Manufacturing and Assembly Model

Rather than investing heavily in manufacturing infrastructure during the early stage, Tejas Labs will adopt a **contract manufacturing and assembly model**.

Under this model, component manufacturing will be handled by specialized suppliers, while final kit assembly will be performed by partner manufacturing units experienced in light assembly operations.

This approach offers several advantages:

- Lower upfront capital investment
- Access to specialized manufacturing expertise
- Operational flexibility to scale production volumes
- Reduced operational risk during early growth stages

The production workflow will follow the following sequence:

- Component manufacturing by specialized suppliers
- Delivery of components to assembly partners
- Assembly of kits according to standardized procedures
- Packaging and labeling of finished products
- Final inspection and quality verification

The assembled kits will then be transported to the company's logistics partner or central distribution hub.

As production volumes grow, Tejas Labs may gradually transition toward semi-integrated manufacturing operations, including dedicated assembly lines and greater production oversight.

6.6 Quality Control and Safety Standards

Ensuring product safety and reliability is critical for a company producing educational products designed for children.

Tejas Labs will implement a multi-stage quality control process covering both component quality and final product performance.

- Quality assurance procedures will include:
- Incoming component inspection to verify supplier quality standards
- Assembly process checks to ensure correct component integration
- Functional testing of completed engineering kits
- Durability testing to ensure repeated use without component failure
- Packaging and instruction verification

Additionally, all kits will include clearly written instructions, safety guidelines, and age recommendations to ensure safe product use.

Maintaining high quality standards will strengthen the brand's credibility among parents, educators, and institutional buyers.

6.7 Infrastructure and Facilities

In the initial operational phase, Tejas Labs will operate with a distributed infrastructure model that minimizes fixed overhead while ensuring operational efficiency.

The operational infrastructure will include three primary components.

6.7.1 Product Development Workspace

A small office or innovation workspace will serve as the center for product design, prototyping, administrative activities, and strategic planning.

This workspace will function as a design studio and innovation lab, where prototypes can be tested and new product ideas can be developed.

6.7.2 Contract Manufacturing Facilities

Manufacturing and assembly operations will be conducted through third-party production partners located in industrial clusters that specialize in light manufacturing and product assembly.

This arrangement allows the company to leverage existing manufacturing expertise while avoiding the cost of establishing its own factory in the early stages.

6.7.3 Warehousing and Logistics Hub

A centralized warehouse will manage finished goods inventory, packaging operations, and order fulfillment.

This warehouse will coordinate directly with logistics providers to ensure efficient nationwide delivery of products.

As order volumes grow, the company may explore the establishment of regional distribution centers to reduce shipping times and logistics costs.

6.8 Technology and Operational Systems

Several technology systems will support the operational efficiency of Tejas Labs.

Product design and engineering will utilize computer-aided design software and 3D modeling tools to enable rapid prototyping and design iteration.

Inventory management software will be implemented to track stock levels, monitor component usage, and support demand forecasting.

Order management systems integrated with the company's e-commerce platform will automate order processing and logistics coordination.

These digital tools allow the company to operate efficiently even with a relatively lean internal team.

6.9 Supply Chain and Logistics Strategy

A resilient supply chain is critical for maintaining consistent product availability and controlling production costs.

Tejas Labs will implement a **multi-tier supply chain strategy** designed to minimize operational disruptions.

Key elements of the supply chain strategy include:

- Diversified supplier networks for critical components
- Strategic procurement planning to prevent stock shortages
- Partnerships with reliable logistics providers for nationwide delivery

For customer deliveries, Tejas Labs will partner with established third-party logistics companies capable of supporting both metropolitan and non-metropolitan delivery networks.

These logistics partnerships allow the company to scale distribution without investing heavily in transportation infrastructure.

6.10 Inventory Management Strategy

Inventory management will follow a **demand-driven production model** designed to balance supply availability with working capital efficiency.

Rather than maintaining large inventories, production will be scheduled in controlled batches aligned with forecasted demand.

Inventory will be managed across three primary categories:

- Raw materials and components received from suppliers
- Work-in-progress kits undergoing assembly
- Finished goods ready for shipment

Digital inventory tracking systems will allow the company to monitor stock levels in real time and optimize production planning.

6.11 Operational Scaling Roadmap

As Tejas Labs grows and achieves consistent sales growth, the company will gradually expand its operational capabilities.

- Key scaling initiatives may include:
- Establishing dedicated assembly facilities to increase production capacity
- Developing long-term supplier contracts to stabilize procurement costs
- Investing in automated assembly and packaging systems
- Building regional distribution hubs to improve logistics efficiency

These improvements will allow Tejas Labs to transition from a lean startup model to a more integrated manufacturing and distribution organization.

6.12 Long-Term Operational Vision

Over the long term, Tejas Labs aims to develop a comprehensive innovation and manufacturing ecosystem for STEM educational products in India.

This vision includes the creation of a dedicated product innovation laboratory where engineers, educators, and designers collaborate to develop new learning experiences and experimental kits.

By integrating product design, educational content, and scalable manufacturing capabilities, Tejas Labs aims to establish itself as a leading provider of experiential STEM learning tools within the Indian education ecosystem.

7. Financial Plan

7.1 Financial Strategy Overview

The financial plan for **Tejas Labs** has been developed to demonstrate the long-term commercial viability of the company and its ability to scale within the growing Indian STEM education market. The financial projections cover a five-year period and include detailed forecasts for revenue growth, operating costs, capital investments, and free cash flow generation.

The company operates at the intersection of educational technology and physical learning products, combining hardware-based STEM kits with structured learning programs. This hybrid business model provides multiple revenue streams, including direct product sales, recurring subscription services, and institutional partnerships with schools.

In the early stages, Tejas Labs will operate as a bootstrapped startup, funded by the founders Dev Vora (Chief Executive Officer) and Siddharth Vaddepalli (Chief Financial Officer). The founders will provide the initial equity capital required for product development, manufacturing setup, and early marketing activities. Limited debt financing may also be used to support working capital and operational scaling.

The financial projections assume that the company will initially focus on direct-to-consumer online distribution, followed by expansion into school partnerships and institutional STEM lab

installations. This approach allows the company to build brand recognition before scaling through larger institutional channels.

As the company grows, operational efficiencies and economies of scale are expected to improve profitability. The financial model reflects gradual improvements in production costs, customer acquisition efficiency, and overall operating margins.

7.2 Key Financial Assumptions

All financial projections are based on a set of realistic operational and economic assumptions. These assumptions are derived from industry benchmarks for early-stage hardware startups, educational product companies, and subscription-based learning platforms.

The projections assume moderate growth during the initial years as the company establishes its brand, followed by accelerated expansion as the product portfolio and distribution network mature.

Core Financial Assumptions

Parameter	Assumption
Corporate Tax Rate	30%
Planning Horizon	5 Years
Inflation Adjustment	5% annually
Depreciation Method	Straight Line
Useful Life of Assets	5 Years
Working Capital Requirement	10% of Revenue
Terminal Growth Rate	4%
Average Gross Margin	60%

These assumptions ensure that the financial model remains conservative while still reflecting the growth potential of the business.

7.3 Initial Capital Investment

The company requires initial capital to fund product design, prototype development, manufacturing setup, and early marketing activities.

The founders will provide the majority of the initial investment, supplemented by a small working capital loan.

7.3.1 Initial Capital Structure

Source of Capital	Amount
Founder Equity Contribution	₹20,00,000
Startup Loan	₹10,00,000
Total Initial Capital	₹30,00,000

This capital will be allocated toward product development, operational setup, and market entry.

7.3.2 Allocation of Initial Investment

Category	Investment
Product Development and Prototyping	₹6,00,000
Manufacturing Equipment	₹12,00,000
Technology Infrastructure	₹7,00,000
Office Equipment and Setup	₹5,00,000
Total Investment	₹30,00,000

These investments represent the core operational infrastructure required to launch the business.

7.4 Revenue Model

Tejas Labs generates revenue through three primary channels. This diversified revenue structure improves financial stability and reduces dependence on a single market segment.

1. Direct Product Sales

The primary revenue stream consists of **STEM learning kits** sold through online marketplaces, the company's website, and educational retail partners.

These kits are designed to provide hands-on learning experiences that allow students to explore engineering concepts through experiments and project-based activities.

2. Subscription Learning Kits

The company will introduce a **monthly STEM experiment subscription program**. Subscribers receive new experimental modules every month along with guided learning materials and digital tutorials.

This recurring revenue stream helps increase customer engagement while improving revenue predictability.

3. Institutional Sales

Educational institutions represent a major long-term growth opportunity. Schools can purchase STEM lab setups and curriculum integration packages that allow them to incorporate hands-on engineering activities into their classrooms.

Institutional contracts typically involve bulk purchases and multi-year partnerships.

7.5 Product Portfolio and Pricing Strategy

The company offers a tiered product portfolio designed to appeal to students with different levels of technical curiosity and experience.

Each kit contains physical components, guided experiment manuals, and access to online learning content.

Product Categories

Product Category	Description	Price
Starter Engineering Kit	Entry-level STEM kit introducing basic engineering concepts	₹1,999

Product Category	Description	Price
Advanced Engineering Kit	Intermediate kit focused on robotics and electronics	₹3,499
Premium Robotics Kit	Advanced programmable robotics platform	₹6,999

This pricing structure is designed to balance affordability with profitability while remaining competitive within the Indian educational product market.

7.6 Sales Forecast Methodology

The sales forecast is based on several market and operational factors, including:

- Growth of the STEM education sector in India
- Increasing adoption of experiential learning methods
- Expansion of online educational marketplaces
- Partnerships with schools and educational institutions

During the first year, sales are expected to come primarily from **online direct-to-consumer channels**. As brand awareness grows, institutional partnerships will begin contributing a larger share of revenue.

The projections assume gradual increases in sales volume as marketing efforts and distribution channels expand.

7.7 Projected Unit Sales

The following table outlines the projected unit sales for each product category over the five-year planning period.

Year	Starter Kits	Advanced Kits	Premium Kits
2026	2,000	800	200
2027	4,000	1,800	400
2028	7,500	3,200	800
2029	12,000	6,000	1,800
2030	18,000	10,000	4,000

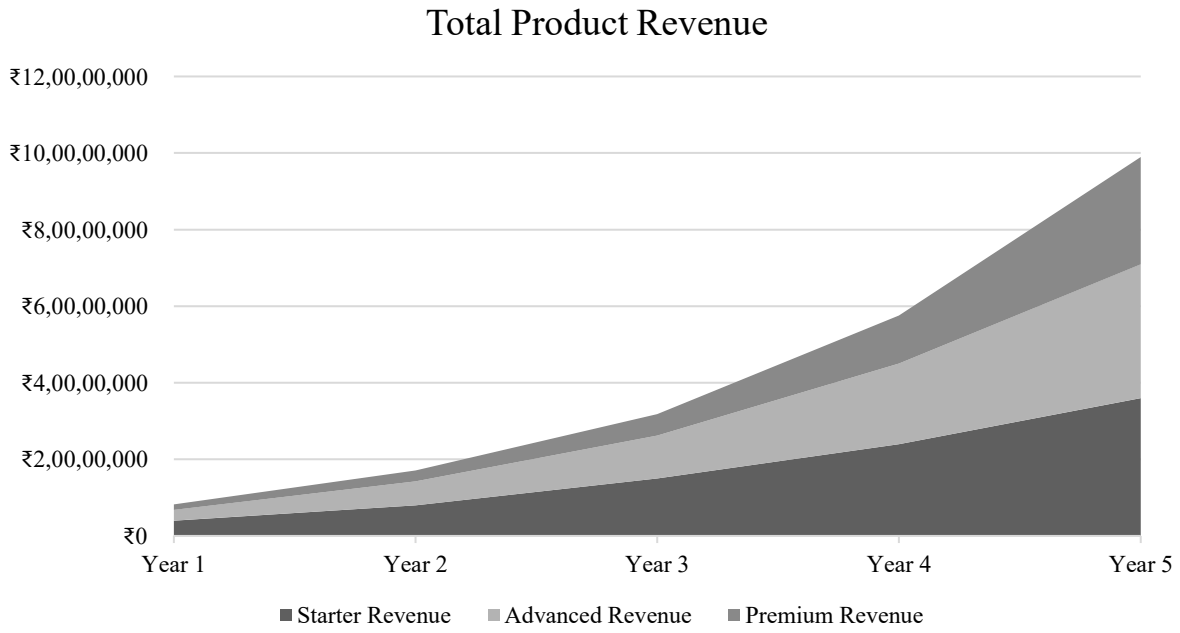
These projections assume that the company gradually expands its customer base through digital marketing, partnerships with educational influencers, and collaborations with schools.

7.8 Product Revenue Projection

Based on the projected sales volumes and pricing structure, the expected product revenue for the first five years is calculated as follows.

Year	Starter Revenue	Advanced Revenue	Premium Revenue	Total Product Revenue
2026	₹39,98,000	₹27,99,200	₹13,99,800	₹81,97,000
2027	₹79,96,000	₹62,98,200	₹27,99,600	₹1,70,93,800

Year	Starter Revenue	Advanced Revenue	Premium Revenue	Total Product Revenue
2028	₹1,49,92,500	₹1,11,96,800	₹55,99,200	₹3,17,88,500
2029	₹2,39,88,000	₹2,09,94,000	₹1,25,98,200	₹5,75,80,200
2030	₹3,59,82,000	₹3,49,90,000	₹2,79,96,000	₹9,89,68,000



These projections reflect the gradual transition from an early-stage startup to a growing educational product company.

7.9 Subscription Revenue Model

While product sales represent the primary entry point into the market, Tejas Labs also intends to build a strong recurring revenue base through its **STEM Experiment Subscription Program**. The subscription model is designed to deepen customer engagement while creating predictable and stable revenue streams.

Under this model, subscribers receive a new experimental module each month that expands on previously learned concepts. Each subscription box includes additional components, structured experiment guides, and access to exclusive digital tutorials and online workshops.

The subscription offering is priced at **₹999 per month**, which positions it competitively within the Indian educational enrichment market while maintaining healthy contribution margins.

Adoption of the subscription model is expected to grow gradually as the company builds a loyal customer base through its core product offerings.

Projected Subscription Growth

Year	Subscribers	Annual Subscription Revenue
2026	300	₹35,96,400
2027	600	₹71,92,800

Year	Subscribers	Annual Subscription Revenue
2028	2,000	₹2,39,76,000
2029	3,500	₹4,19,58,000
2030	6,000	₹7,19,28,000

The growth in subscription adoption is expected to be driven by:

- positive customer experiences with the initial kits
- continuous product innovation
- community-driven STEM learning initiatives

Over time, the subscription service is expected to become one of the company's most valuable revenue streams due to its recurring nature.

7.10 Institutional Revenue Model

Educational institutions represent a significant long-term growth opportunity for Tejas Labs. Schools across India are increasingly incorporating **experiential learning and STEM laboratories** into their curriculum, creating demand for structured hands-on learning solutions.

Tejas Labs plans to partner with schools by offering:

- STEM laboratory setup packages
- classroom learning kits
- teacher training modules
- curriculum integration support

Institutional contracts typically involve bulk purchases and can generate significantly larger revenue per customer compared to individual sales.

Projected Institutional Partnerships

Year	Partner Schools	Revenue from Schools
2026	5	₹75,00,000
2027	8	₹1,20,00,000
2028	12	₹1,80,00,000
2029	20	₹3,00,00,000
2030	50	₹7,50,00,000

These partnerships are expected to emerge gradually as the company builds credibility in the education sector.

Institutional sales are particularly attractive because they generate:

- higher average order values
- stable long-term relationships
- predictable demand cycles

7.11 Consolidated Revenue Projection

By combining all revenue streams, Tejas Labs is expected to achieve strong revenue growth over the five-year planning period.

The diversified revenue model helps reduce financial risk while allowing the company to scale across multiple market segments.

Total Revenue Projection

Year	Product Revenue	Subscription Revenue	Institutional Revenue	Total Revenue
2026	₹0.82 Cr	₹0.36 Cr	₹0.75 Cr	**₹1.93 Cr**
2027	₹1.71 Cr	₹0.72 Cr	₹1.20 Cr	**₹3.63 Cr**
2028	₹3.18 Cr	₹2.40 Cr	₹1.80 Cr	**₹7.38 Cr**
2029	₹5.76 Cr	₹4.20 Cr	₹3.00 Cr	**₹12.96 Cr**
2030	₹9.90 Cr	₹7.20 Cr	₹7.50 Cr	**₹24.60 Cr**

The projections suggest that revenue will grow at a strong compounded rate as the company expands its distribution channels and strengthens its brand presence.

7.12 Cost Structure

The overall cost structure of Tejas Labs consists of two major categories: **variable costs** and **fixed operating costs**. Understanding this distinction is important for evaluating the scalability and profitability of the business model.

Variable costs increase in proportion to sales volume because they are directly associated with the production and delivery of the STEM kits. Fixed operating costs, on the other hand, represent expenses required to run the business regardless of the number of products sold.

The cost structure has been designed to maintain **high gross margins**, which is typical for education-oriented hardware products where intellectual property, curriculum design, and brand value play an important role.

7.12.1 Variable Costs

Variable costs represent the direct cost of producing and delivering each product sold. These costs scale with the number of units sold and therefore increase as the company grows.

For Tejas Labs, the primary components of variable costs include:

- electronic components and mechanical parts used in STEM kits
- manufacturing and assembly costs
- packaging materials
- logistics and shipping expenses
- payment gateway and marketplace commissions

Based on industry benchmarks and estimated bill of materials, the **average variable cost is assumed to be approximately 40% of total revenue**.

7.12.1.1 Variable Cost Breakdown

Cost Component	% of Revenue
Electronic components and raw materials	22%
Manufacturing and assembly	8%
Packaging	4%

Cost Component	% of Revenue
Shipping and logistics	4%
Payment gateway / marketplace fees	2%
Total Variable Cost	**40%**

Applying this structure to projected revenues produces the following cost estimates.

7.12.1.2 Projected Variable Costs

Year	Revenue	Variable Cost (40%)
2026	₹1.93 Cr	₹0.77 Cr
2027	₹3.63 Cr	₹1.45 Cr
2028	₹7.38 Cr	₹2.95 Cr
2029	₹12.96 Cr	₹5.18 Cr
2030	₹24.60 Cr	₹9.84 Cr

These costs correspond directly to the **Cost of Goods Sold (COGS)** in the projected income statement.

7.12.2 Fixed Operating Costs

Fixed operating costs include expenses necessary to maintain the company's operations and support business growth. These costs do not change significantly with short-term changes in production volume.

As the company scales, some fixed costs such as marketing and employee salaries will gradually increase to support expansion.

The major components of fixed operating costs include personnel expenses, marketing investments, research and development activities, and administrative overhead.

7.12.2.1 Fixed Cost Categories

Expense Category	Description
Employee Salaries	Engineering, operations, marketing staff
Marketing and Brand Development	Digital marketing, campaigns, influencer partnerships
Research and Development	Product design and new experiment development
Office Rent and Utilities	Workspace costs
Technology Infrastructure	Cloud services, software tools
Administrative Expenses	Legal, accounting, compliance

7.12.2.2 Projected Fixed Operating Expenses

The following table outlines the estimated fixed operating expenses for each year of operation.

Expense Category	2026	2027	2028	2029	2030
Employee Salaries	₹30 L	₹45 L	₹80 L	₹130 L	₹200 L
Marketing and Customer Acquisition	₹20 L	₹35 L	₹60 L	₹100 L	₹160 L
Research and Development	₹15 L	₹20 L	₹35 L	₹60 L	₹80 L
Office Rent and Utilities	₹10 L	₹12 L	₹15 L	₹20 L	₹25 L
Technology Infrastructure	₹8 L	₹10 L	₹15 L	₹20 L	₹30 L
Administrative and Legal	₹7 L	₹13 L	₹35 L	₹80 L	₹155 L
Total Operating Expenses	**₹90 L**	**₹135 L**	**₹240 L**	**₹410 L**	**₹650 L**

These operating costs align with the **Operating Expense line in the Profit and Loss statement**.

7.13 Capital Expenditure and Depreciation

To support product manufacturing and operational activities, the company must invest in essential infrastructure and development facilities.

7.13.1 Initial Capital Expenditure

Asset Category	Investment
Manufacturing Tools and Equipment	₹12,00,000
Prototype Development Lab	₹6,00,000
Technology Infrastructure	₹7,00,000
Office Equipment	₹5,00,000
Total Capital Expenditure	**₹30,00,000**

These assets are assumed to have an economic life of approximately five years.

Using the straight-line depreciation method, the annual depreciation expense is calculated as follows:

Annual Depreciation = Total Capital Expenditure ÷ Useful Life

Annual Depreciation = ₹30,00,000 ÷ 5

Annual Depreciation = **₹6,00,000 per year**

Depreciation is a non-cash accounting expense that reduces taxable income while reflecting the gradual consumption of productive assets.

7.14 Projected Profit and Loss Statement

The projected income statement summarizes the financial performance of the company based on the revenue projections and cost assumptions discussed earlier.

7.14.1 Projected Profit and Loss Statement

Year	2026	2027	2028	2029	2030
Revenue	1.93 Cr	3.63 Cr	7.38 Cr	12.96 Cr	24.60 Cr
Cost of Goods Sold (Variable Costs)	0.77 Cr	1.45 Cr	2.95 Cr	5.18 Cr	9.84 Cr
Gross Profit	1.16 Cr	2.18 Cr	4.43 Cr	7.78 Cr	14.76 Cr
Operating Expenses (Fixed Costs)	0.90 Cr	1.35 Cr	2.40 Cr	4.10 Cr	6.50 Cr
EBITDA	0.26 Cr	0.83 Cr	2.03 Cr	3.68 Cr	8.26 Cr
Depreciation	0.06 Cr	0.06 Cr	0.06 Cr	0.06 Cr	0.06 Cr
EBIT	0.20 Cr	0.77 Cr	1.97 Cr	3.62 Cr	8.20 Cr
Interest Expense	0.10 Cr	0.08 Cr	0.06 Cr	0.04 Cr	0
Profit Before Tax	0.10 Cr	0.69 Cr	1.91 Cr	3.58 Cr	8.20 Cr
Tax (30%)	0.03 Cr	0.21 Cr	0.57 Cr	1.07 Cr	2.46 Cr
Net Profit	**0.07 Cr**	**0.48 Cr**	**1.34 Cr**	**2.51 Cr**	**5.74 Cr**

The projections indicate that Tejas Labs becomes profitable early due to strong gross margins and scalable operations.

7.15 Working Capital Requirements

Working capital represents the short-term capital required to support the day-to-day operations of the business. For a product-oriented company such as Tejas Labs, working capital primarily consists of inventory, accounts receivable, and operational cash balances required to maintain uninterrupted business activities.

As the company scales its operations, additional working capital will be required to support higher levels of inventory, increased production volumes, and expanding sales. Educational kits require a steady supply of electronic components and materials, which necessitates maintaining sufficient inventory buffers. In addition, institutional sales to schools and educational institutions may involve short credit cycles, leading to receivable balances.

To reflect these operational needs, the financial model assumes that working capital requirements will be approximately **10 percent of annual revenue**. This assumption is consistent with early-stage manufacturing and product-based companies where inventory and receivables form a moderate portion of operating capital.

7.15.1 Projected Working Capital Requirement

Year	Revenue	Working Capital (10%)
2026	₹1.93 Cr	₹0.19 Cr
2027	₹3.63 Cr	₹0.36 Cr
2028	₹7.38 Cr	₹0.74 Cr
2029	₹12.96 Cr	₹1.30 Cr

Year	Revenue	Working Capital (10%)
2030	₹24.60 Cr	₹2.46 Cr

For cash flow calculations, the relevant measure is the **change in working capital**, which represents the incremental investment required each year as the business grows.

7.15.2 Change in Working Capital

Year	Working Capital	Increase in Working Capital
2026	0.19 Cr	0.19 Cr
2027	0.36 Cr	0.17 Cr
2028	0.74 Cr	0.38 Cr
2029	1.30 Cr	0.56 Cr
2030	2.46 Cr	1.16 Cr

An increase in working capital represents a **cash outflow**, since additional funds must be invested in operational activities to support growth.

7.16 Debt Financing and Repayment Structure

To support early operational requirements, Tejas Labs will utilize a modest amount of external financing in the form of a term loan. The purpose of this financing is to provide liquidity for working capital and early-stage operational expenses while maintaining financial flexibility.

The loan structure assumed in this financial model is summarized below.

7.16.1 Loan Assumptions

Parameter	Value
Loan Amount	₹10,00,000
Interest Rate	10% per annum
Loan Tenure	5 Years
Repayment Method	Equal Principal Repayment

Under this structure, the company repays an equal portion of principal each year while interest is calculated on the outstanding loan balance.

7.16.2 Debt Repayment Schedule

Year	Opening Balance	Principal Repayment	Interest	Closing Balance
2026	10 L	2 L	1.0 L	8 L
2027	8 L	2 L	0.8 L	6 L
2028	6 L	2 L	0.6 L	4 L
2029	4 L	2 L	0.4 L	2 L

Year	Opening Balance	Principal Repayment	Interest	Closing Balance
2030	2 L	2 L	0.2 L	0

The interest expense calculated above is reflected in the **profit and loss statement**, while principal repayments are incorporated into the **equity cash flow calculations**.

7.17 Free Cash Flow to the Firm (FCFF)

Free Cash Flow to the Firm (FCFF) represents the cash generated by the company that is available to all capital providers, including both equity investors and debt holders. Unlike accounting profit, FCFF measures the actual cash generated by the business after accounting for operational reinvestments.

FCFF is widely used in financial valuation because it reflects the economic cash-generating capability of the firm, independent of its financing structure.

The FCFF calculation is based on the following formula:

$$FCFF = EBIT \times (1 - Tax\ Rate) + Depreciation - Capital\ Expenditure - Increase\ in\ Working\ Capital$$

Each component of this formula captures a key financial adjustment:

EBIT × (1 − Tax Rate) represents **Net Operating Profit After Tax (NOPAT)**, which reflects the after-tax operating earnings generated by the business.

Depreciation is added back because it is a non-cash accounting expense that reduces reported profit but does not involve an actual cash outflow.

Capital expenditure represents investments in long-term productive assets such as manufacturing tools, development equipment, and infrastructure required for operational growth.

The increase in working capital represents additional operating investment needed to support higher production and sales.

7.17.1 NOPAT Calculation

TE JAS LABS

Year	EBIT	Tax Rate	NOPAT
2026	0.20 Cr	30%	0.14 Cr
2027	0.77 Cr	30%	0.54 Cr
2028	1.97 Cr	30%	1.38 Cr
2029	3.62 Cr	30%	2.53 Cr
2030	8.20 Cr	30%	5.74 Cr

7.17.2 Free Cash Flow to the Firm Calculation

Component	2026	2027	2028	2029	2030
NOPAT	0.14	0.54	1.38	2.53	5.74
Depreciation	0.06	0.06	0.06	0.06	0.06
Capital Expenditure	(0.30)	(0.10)	(0.10)	(0.10)	(0.10)

Component	2026	2027	2028	2029	2030
Increase in Working Capital	(0.19)	(0.17)	(0.38)	(0.56)	(1.16)
FCFF	**-0.29**	**0.33**	**0.96**	**1.93**	**4.54**

The negative FCFF in the first year reflects the initial investment phase where capital expenditures and working capital investments exceed operating cash inflows. As the business scales, operating cash generation improves significantly, resulting in strong positive free cash flows in subsequent years.

7.18 Free Cash Flow to Equity (FCFE)

Free Cash Flow to Equity represents the cash flow available to equity shareholders after accounting for operational investments and debt obligations. This metric is particularly important for evaluating the potential return available to founders and investors.

FCFE incorporates the impact of financing decisions, including borrowing and debt repayments.

The FCFE formula is as follows:

$$FCFE = Net\ Income + Depreciation - Capital\ Expenditure - Increase\ in\ Working\ Capital + Net\ Borrowing$$

Where Net Borrowing represents new debt raised minus principal repayments.

7.18.1 FCFE Calculation

Component	2026	2027	2028	2029	2030
Net Income	0.07	0.48	1.34	2.51	5.74
Depreciation	0.06	0.06	0.06	0.06	0.06
Capital Expenditure	(0.30)	(0.10)	(0.10)	(0.10)	(0.10)
Increase in WC	(0.19)	(0.17)	(0.38)	(0.56)	(1.16)
Net Borrowing	0.10	(0.02)	(0.02)	(0.02)	(0.02)
FCFE	**-0.26**	**0.25**	**0.90**	**1.89**	**4.52**

The negative FCFE in the first year reflects the startup phase where significant investments are made to build operational capacity. As revenues grow and capital investments stabilize, the company begins generating substantial cash flows available to equity investors.

7.19 Cost of Equity (CAPM)

The cost of equity represents the return expected by equity investors for investing capital in the company. Since equity investors bear the residual risk of the business, this required return reflects both market risk and company-specific risk.

The cost of equity is estimated using the **Capital Asset Pricing Model (CAPM)**. This model calculates the expected return on equity based on the risk-free rate, the market risk premium, and the systematic risk (beta) of the company.

The formula for CAPM is:

$$Cost\ of\ Equity = Risk\ Free\ Rate + Beta \times (Market\ Return - Risk\ Free\ Rate)$$

For this financial model, the following assumptions are used.

Parameter	Assumption
Risk-Free Rate	7%
Market Return	14%
Equity Beta	1.2

The risk-free rate is approximated using long-term Indian government bond yields. The market return represents the expected return from the equity market, while beta measures the sensitivity of the firm's returns relative to market movements.

Using the CAPM formula:

Cost of Equity = 7% + 1.2 × (14% − 7%)

Cost of Equity = 7% + 1.2 × 7%

Cost of Equity = **15.4%**

Thus, the expected return required by equity investors is estimated at **15.4 percent**.

7.20 Cost of Debt

The cost of debt represents the effective interest rate paid on borrowed capital. Debt financing provides a tax advantage because interest expenses are tax deductible.

The after-tax cost of debt is calculated using the formula:

$$After - Tax\ Cost\ of\ Debt\ =\ Interest\ Rate\ \times\ (1\ -\ Tax\ Rate)$$

Assuming a loan interest rate of **10 percent** and a corporate tax rate of **30 percent**, the cost of debt becomes:

After-Tax Cost of Debt = 10% × (1 − 0.30)

After-Tax Cost of Debt = **7%**

7.21 Weighted Average Cost of Capital (WACC)

The Weighted Average Cost of Capital represents the overall cost of financing the firm's operations using both equity and debt. WACC is used as the discount rate for valuing the firm's free cash flows.

The formula for WACC is:

$$WACC\ =\ (E\ /\ V\ \times\ Re)\ +\ (D\ /\ V\ \times\ Rd\ \times\ (1\ -\ T))$$

Where:

$$E\ =\ Market\ value\ of\ equity$$
$$D\ =\ Market\ value\ of\ debt$$
$$V\ =\ Total\ firm\ value\ (E\ +\ D)$$
$$Re\ =\ Cost\ of\ equity$$
$$Rd\ =\ Cost\ of\ debt$$
$$T\ =\ Corporate\ tax\ rate$$

For the purposes of this financial model, the company is assumed to maintain the following capital structure.

Capital Component	Value
Equity	80%
Debt	20%

Substituting the values:

WACC = (0.80 × 15.4%) + (0.20 × 7%)

WACC = 12.32% + 1.40%

WACC = **13.72%**

Thus, the company's weighted average cost of capital is estimated at **13.72 percent**.

7.22 Discounted Cash Flow (DCF) Valuation

The Discounted Cash Flow method estimates the intrinsic value of the firm by discounting projected free cash flows using the weighted average cost of capital.

The projected **Free Cash Flow to Firm (FCFF)** values calculated earlier are used for this valuation.

7.22.1 FCFF Projections

Year	FCFF (Cr)
2026	-0.29
2027	0.33
2028	0.96
2029	1.93
2030	4.54

These cash flows are discounted using the calculated WACC of **13.72 percent**.

7.22.2 Present Value of FCFF

Year	FCFF	Discount Factor	Present Value
2026	-0.29	0.88	-0.26
2027	0.33	0.77	0.25
2028	0.96	0.68	0.65
2029	1.93	0.60	1.16
2030	4.54	0.53	2.41

Total Present Value of FCFF = **₹4.21 Cr**

7.23 Terminal Value

Since companies operate indefinitely, valuation models include a **terminal value** to capture the value of cash flows beyond the explicit forecast period.

The terminal value is estimated using the **Gordon Growth Model**.

Terminal Value Formula:

$$Terminal\ Value\ =\ FCFF_5\ \times\ (1\ +\ g)\ /\ (WACC\ -\ g)$$

Where:

g = long-term growth rate

Assuming a conservative long-term growth rate of **5 percent**, the terminal value becomes:

Terminal Value = 4.54 × (1.05) / (0.1372 − 0.05)

Terminal Value ≈ **₹54.65 Cr**

The present value of the terminal value is calculated by discounting it back to present value.

Present Value of Terminal Value = 54.65 × 0.53 ≈ **₹28.96 Cr**

7.24 Enterprise Value Estimation

The enterprise value is obtained by summing the present value of projected FCFF and the present value of the terminal value.

$$Enterprise\ Value\ =\ PV\ of\ FCFF\ +\ PV\ of\ Terminal\ Value$$

Enterprise Value = 4.21 + 28.96

Enterprise Value = **₹33.17 Cr**

To obtain the equity value, outstanding debt is subtracted.

Equity Value = Enterprise Value − Debt

Equity Value = 33.17 − 0.10

Equity Value ≈ **₹33.07 Cr**

7.25 Break-Even Analysis

Break-even analysis identifies the level of sales required for the company to cover all operating costs without generating profit or loss.

The break-even revenue is calculated using the formula:

$$Break-Even\ Revenue\ =\ Fixed\ Costs\ /\ Contribution\ Margin$$

Where:

$$Contribution\ Margin\ =\ (Revenue\ -\ Variable\ Cost)\ /\ Revenue$$

Given that variable costs represent **40 percent of revenue**, the contribution margin becomes **60 percent**.

Using 2026 fixed operating costs:

Break-Even Revenue = 0.90 Cr / 0.60

Break-Even Revenue = **₹1.50 Cr**

Since projected revenue in 2026 is ₹1.93 Cr, the company is expected to **cross the break-even point during the first year of operations**.

7.26 Key Financial Ratios

Financial ratios help evaluate the profitability, efficiency, and sustainability of the business model.

7.26.1 Profitability Ratios

Ratio	2026	2030
Gross Margin	60%	60%
EBIT Margin	10%	33%
Net Profit Margin	3.6%	23%

7.26.2 Efficiency Ratios

Ratio	2026	2030
Revenue Growth	—	90%
Asset Turnover	1.8	4.5

These ratios indicate strong scalability, with margins improving significantly as the business expands.

8. Conclusion

The financial plan developed for Tejas Labs demonstrates the viability and long-term potential of the proposed business model. Through structured revenue projections, cost analysis, and valuation techniques, the analysis provides a comprehensive understanding of the company's expected financial performance over the first five years of operation.

The projected income statement indicates steady revenue growth driven by increasing adoption of STEM education kits and expansion into institutional and direct-to-consumer channels. With a variable cost structure representing approximately 40 percent of revenue, the business is able to maintain a healthy gross margin of nearly 60 percent. This strong margin structure supports the company's ability to cover operating expenses while investing in marketing, research and development, and organizational expansion.

Although the company experiences modest cash flow pressure during the early phase due to capital expenditures and working capital investments, the financial projections show that operating cash flows improve significantly as the business scales. The Free Cash Flow to the Firm analysis highlights a transition from initial investment-driven negative cash flows to strong positive cash generation in the later years. Similarly, the Free Cash Flow to Equity projections indicate increasing returns available to shareholders as operational efficiencies improve and revenue grows.

The valuation analysis conducted using the Discounted Cash Flow method estimates an enterprise value of approximately ₹33 crore, reflecting the long-term economic value created by the business. The break-even analysis further indicates that the firm can achieve operational sustainability relatively early, with projected revenues exceeding the break-even threshold within the first year of operations.

Overall, the financial projections suggest that Tejas Labs possesses a scalable and financially sustainable business model. The combination of strong gross margins, improving profitability, and increasing cash flow generation indicates significant growth potential in the expanding STEM education market. While the projections rely on certain assumptions regarding market demand, cost structures, and growth rates, the overall analysis demonstrates that the venture has the potential to generate meaningful economic value for both founders and investors.

In conclusion, the financial plan supports the strategic vision of Tejas Labs as a technology-enabled education company capable of delivering innovative learning products while maintaining financial sustainability and long-term growth prospects.